Week 4: Properties of Estimators

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Introduction to Data Science (MSc)

1 Introduction to Estimator Properties

The theory of estimation forms the mathematical foundation of statistical inference and data science methodology, providing rigorous criteria for evaluating and comparing different approaches to parameter estimation. Understanding the fundamental properties of estimators—bias, variance, consistency, efficiency, and robustness—is essential for developing reliable data analysis methods and interpreting their results correctly.

The importance of estimator properties extends far beyond theoretical considerations. In modern data science applications, where datasets are often large, high-dimensional, and potentially contaminated with outliers or measurement errors, the choice of estimation method can dramatically affect the reliability and interpretability of results. The mathematical framework developed in this lecture provides the tools necessary to make informed decisions about estimation procedures and to understand their limitations.

From a practical perspective, the properties studied here directly inform the design of machine learning algorithms, the selection of statistical models, and the interpretation of uncertainty in data-driven conclusions. The bias-variance tradeoff, in particular, underlies many fundamental concepts in machine learning, including regularization, model selection, and ensemble methods.

2 Mathematical Framework for Bias and Variance

2.1 Bias: Systematic Deviations from Truth

The concept of bias captures the systematic tendency of an estimator to deviate from the true parameter value. This deviation represents a fundamental limitation that cannot be reduced by increasing sample size, making bias a critical consideration in estimator evaluation.

Definition 2.1 (Bias of an Estimator)**.** *Let $\hat{\theta}_n$ be an estimator of parameter θ based on a sample of size n. The bias of $\hat{\theta}_n$ is defined as:*

$$Bias(\hat{\theta}_n) = E_\theta[\hat{\theta}_n] - \theta$$

where the expectation is taken with respect to the sampling distribution when the true parameter value is θ.

The subscript θ in E_θ emphasizes that the expectation depends on the true parameter value, highlighting that bias may vary across the parameter space. An estimator may be unbiased for some parameter values while exhibiting bias for others, making the uniform evaluation of bias across the entire parameter space important for theoretical analysis.

Definition 2.2 (Unbiased Estimator)**.** *An estimator $\hat{\theta}_n$ is unbiased for θ if:*

$$E_\theta[\hat{\theta}_n] = \theta \quad \textit{for all } \theta \in \Theta$$

where Θ is the parameter space.

The requirement that unbiasedness hold for all values in the parameter space is stringent and may not always be achievable or even desirable. In many practical situations, a small amount of bias may be acceptable if it leads to substantial reductions in variance, as formalized in the bias-variance tradeoff.

Theorem 2.1 (Unbiasedness of Sample Moments)**.** *Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables with finite moments. Then:*

1. *The sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ is an unbiased estimator of $\mu = E[X_i]$*
2. *The sample variance $S_n^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ is an unbiased estimator of $\sigma^2 = Var(X_i)$*

Proof. For the sample mean:

$$E[\bar{X}_n] = E\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n}\sum_{i=1}^n E[X_i] = \frac{1}{n}\cdot n\mu = \mu$$

For the sample variance, we use the identity $\sum_{i=1}^n (X_i - \bar{X}_n)^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X}_n - \mu)^2$:

$$\begin{aligned}
E[S_n^2] &= \frac{1}{n-1}E\left[\sum_{i=1}^n (X_i - \bar{X}_n)^2\right] && (1)\\
&= \frac{1}{n-1}E\left[\sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X}_n - \mu)^2\right] && (2)\\
&= \frac{1}{n-1}\left[n\sigma^2 - n\cdot\frac{\sigma^2}{n}\right] = \frac{1}{n-1}(n-1)\sigma^2 = \sigma^2 && (3)
\end{aligned}$$

□

2.2 Variance and the Precision of Estimation

Variance quantifies the variability of an estimator around its expected value, providing a measure of the precision or reliability of the estimation procedure. Unlike bias, variance can typically be reduced by increasing sample size, though the rate of reduction depends on the specific estimator and underlying distribution.

Definition 2.3 (Variance of an Estimator)**.** *The variance of an estimator $\hat{\theta}_n$ is:*

$$Var(\hat{\theta}_n) = E_\theta[(\hat{\theta}_n - E_\theta[\hat{\theta}_n])^2]$$

The variance provides information about the spread of the sampling distribution around its center. Small variance indicates that repeated applications of the estimator to different samples from the same population will yield similar results, while large variance suggests high sensitivity to particular sample realizations.

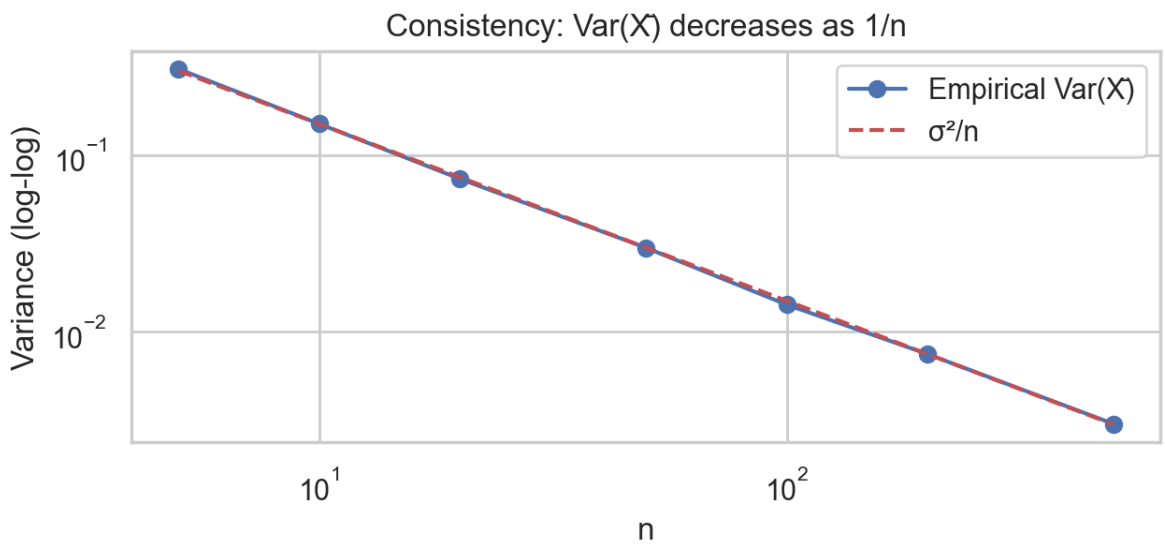

Figure 1: Empirical variance of the sample mean across n compared to the theoretical σ^2/n on log-log scales.

Theorem 2.2 (Variance of Sample Mean)**.** *For independent and identically distributed random variables $X_1, \ldots, X_n$ with finite variance σ^2:*

$$Var(\bar{X}_n) = \frac{\sigma^2}{n}$$

This fundamental result shows that the variance of the sample mean decreases linearly with sample size, providing the theoretical foundation for the intuition that larger samples yield more precise estimates. The $1/n$ rate of decrease is optimal for estimating the mean under mild regularity conditions.

2.3 Mean Squared Error and the Bias-Variance Decomposition

The mean squared error provides a unified framework for evaluating estimator performance by combining both bias and variance into a single measure of estimation quality.

Definition 2.4 (Mean Squared Error)**.** *The mean squared error (MSE) of an estimator $\hat{\theta}_n$ is:*

$$MSE(\hat{\theta}_n) = E_\theta[(\hat{\theta}_n - \theta)^2]$$

The MSE measures the expected squared deviation of the estimator from the true parameter value, providing a natural loss function for comparing estimators. The choice of squared loss is mathematically convenient and corresponds to many practical situations where large errors are disproportionately costly.

Theorem 2.3 (Bias-Variance Decomposition)**.** *The mean squared error can be decomposed as:*

$$MSE(\hat{\theta}_n) = Var(\hat{\theta}_n) + [Bias(\hat{\theta}_n)]^2$$

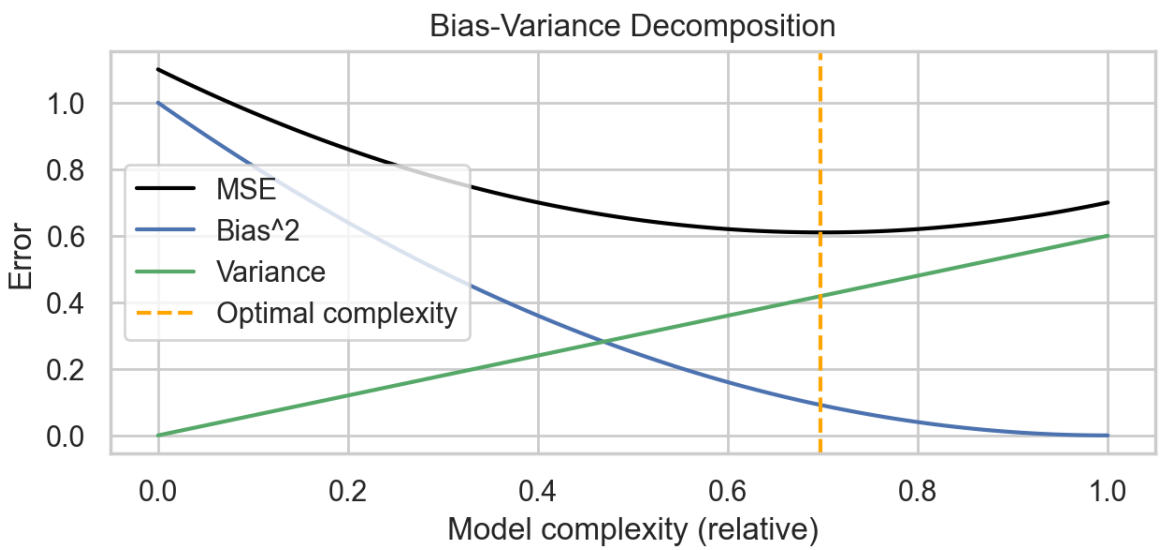

Figure 2: MSE decomposition into bias^2 and variance with optimal complexity indicated.

Proof. Let $\mu_{\hat{\theta}} = E_\theta[\hat{\theta}_n]$ denote the expected value of the estimator. Then:

$$\begin{aligned}
\mathrm{MSE}(\hat{\theta}_n) &= E_\theta[(\hat{\theta}_n - \theta)^2] && (4)\\
&= E_\theta[(\hat{\theta}_n - \mu_{\hat{\theta}} + \mu_{\hat{\theta}} - \theta)^2] && (5)\\
&= E_\theta[(\hat{\theta}_n - \mu_{\hat{\theta}})^2] + 2E_\theta[(\hat{\theta}_n - \mu_{\hat{\theta}})(\mu_{\hat{\theta}} - \theta)] + (\mu_{\hat{\theta}} - \theta)^2 && (6)\\
&= \mathrm{Var}(\hat{\theta}_n) + 0 + [\mathrm{Bias}(\hat{\theta}_n)]^2 && (7)
\end{aligned}$$

where the cross-term vanishes because $E_\theta[\hat{\theta}_n - \mu_{\hat{\theta}}] = 0$. □

This decomposition reveals that the total error of an estimator arises from two distinct sources: the inherent variability of the estimator (variance) and systematic deviations from the true value (bias squared). The decomposition provides crucial insights into the tradeoffs involved in estimator design and forms the theoretical foundation for many machine learning concepts.

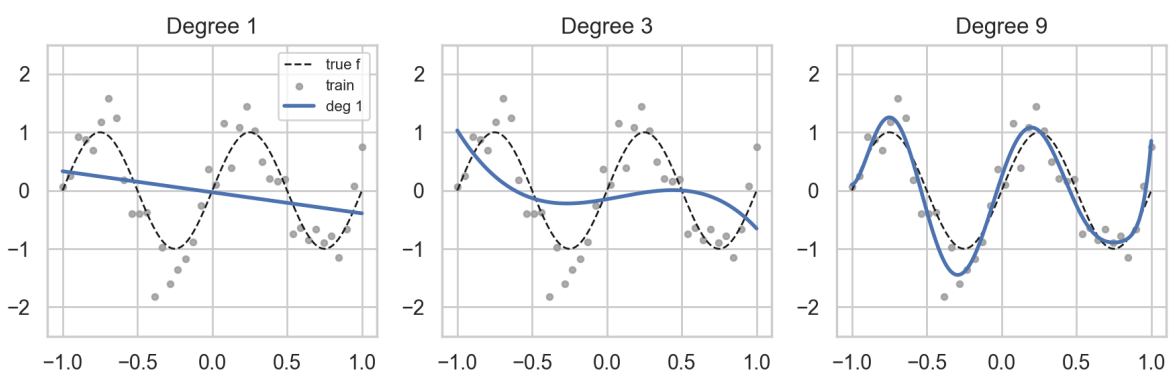

Figure 3: Bias-variance tradeoff illustrated via polynomial regression fits of varying degree to noisy data.

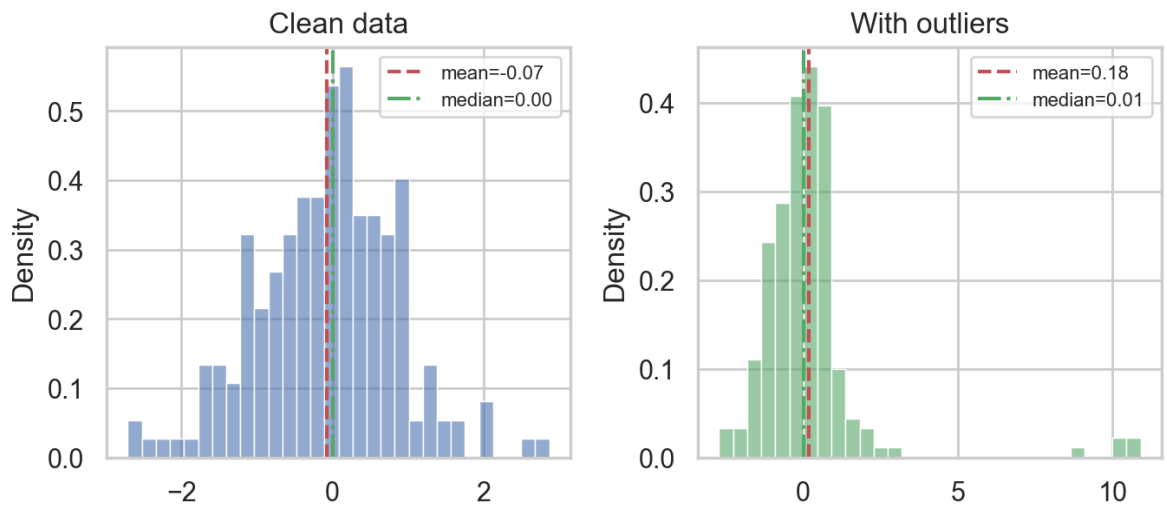

Figure 4: Effect of outliers on mean vs median: the median is less sensitive to extreme values.

3 Consistency: Asymptotic Convergence to Truth

3.1 Mathematical Definition and Types of Convergence

Consistency addresses the fundamental question of whether an estimator approaches the true parameter value as the sample size increases. This asymptotic property is crucial for understanding the long-run behavior of estimation procedures and provides theoretical justification for their use in practice.

Definition 3.1 (Consistency in Probability)**.** *An estimator sequence $\{\hat{\theta}_n\}$ is consistent for θ if:*

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \to \infty$$

where $\xrightarrow{P}$ denotes convergence in probability.

Convergence in probability means that for any $\epsilon > 0$:

$$\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$$

This definition captures the intuitive notion that the estimator becomes increasingly likely to be close to the true parameter value as the sample size grows, though it does not guarantee that any particular realization will be close to the truth.

Definition 3.2 (Strong Consistency)**.** *An estimator sequence $\{\hat{\theta}_n\}$ is strongly consistent (or almost surely consistent) for θ if:*

$$\hat{\theta}_n \xrightarrow{a.s.} \theta \quad \text{as } n \to \infty$$

where $\xrightarrow{a.s.}$ denotes almost sure convergence.

Strong consistency is a stronger condition than consistency in probability, requiring that the estimator converges to the true value with probability one. While this is a desirable property, consistency in probability is often sufficient for practical purposes and is easier to establish theoretically.

Theorem 3.1 (Sufficient Conditions for Consistency)**.** *If* $\lim_{n\to\infty} MSE(\hat{\theta}_n) = 0$*, then* $\hat{\theta}_n$ *is consistent for* θ*.*

Proof. By Markov's inequality, for any $\epsilon > 0$:

$$P(|\hat{\theta}_n - \theta| > \epsilon) \leq \frac{E[(\hat{\theta}_n - \theta)^2]}{\epsilon^2} = \frac{\text{MSE}(\hat{\theta}_n)}{\epsilon^2}$$

If $\lim_{n\to\infty} \text{MSE}(\hat{\theta}_n) = 0$, then $\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$. □

Corollary 3.1 (Consistency via Bias and Variance)**.** *If* $\lim_{n\to\infty} Bias(\hat{\theta}_n) = 0$ *and* $\lim_{n\to\infty} Var(\hat{\theta}_n) = 0$*, then* $\hat{\theta}_n$ *is consistent for* θ*.*

This corollary follows immediately from the bias-variance decomposition and provides a practical approach to establishing consistency by showing that both bias and variance vanish asymptotically.

3.2 Consistency of Common Estimators

Theorem 3.2 (Strong Law of Large Numbers for Sample Mean)**.** *Let* $X_1, X_2, \ldots$ *be independent and identically distributed random variables with finite mean* μ*. Then:*

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{a.s.} \mu \quad \text{as } n \to \infty$$

This fundamental result establishes the strong consistency of the sample mean under minimal conditions, providing theoretical justification for one of the most basic estimation procedures in statistics.

Theorem 3.3 (Consistency of Method of Moments Estimators)**.** *Under regularity conditions, method of moments estimators are consistent. Specifically, if* $\hat{\theta}_n$ *is obtained by solving:*

$$\frac{1}{n}\sum_{i=1}^{n} g(X_i) = E_\theta[g(X)]$$

for appropriate functions g*, then* $\hat{\theta}_n \xrightarrow{P} \theta$*.*

The consistency of method of moments estimators follows from the law of large numbers applied to the sample moments and the continuous mapping theorem, which preserves convergence under continuous transformations.

4 Efficiency and Information-Theoretic Bounds

4.1 Fisher Information and the Cramér-Rao Lower Bound

The concept of efficiency addresses the question of how well an estimator performs relative to theoretical limits on estimation accuracy. The Cramér-Rao lower bound provides a fundamental limit on the variance of unbiased estimators, establishing a benchmark for evaluating estimator performance.

Definition 4.1 (Fisher Information)**.** *For a parameter θ and probability density function $f(x;\theta)$, the Fisher information is:*

$$I(\theta) = E_\theta\left[\left(\frac{\partial \log f(X;\theta)}{\partial\theta}\right)^2\right]$$

provided the expectation exists and is finite.

Under regularity conditions that allow differentiation and integration to be interchanged, the Fisher information can also be expressed as:

$$I(\theta) = -E_\theta\left[\frac{\partial^2 \log f(X;\theta)}{\partial\theta^2}\right]$$

The Fisher information quantifies the amount of information about θ contained in a single observation. Larger values indicate that the data are more informative about the parameter, leading to more precise estimation.

Theorem 4.1 (Cramér-Rao Lower Bound)**.** *Let $\hat{\theta}$ be an unbiased estimator of θ based on a sample of size n. Under regularity conditions:*

$$Var(\hat{\theta}) \geq \frac{1}{nI(\theta)}$$

where $I(\theta)$ is the Fisher information for a single observation.

The Cramér-Rao bound establishes a fundamental limit on the precision of unbiased estimation, showing that no unbiased estimator can have variance smaller than the inverse of the Fisher information. This bound is achieved by maximum likelihood estimators under regularity conditions, establishing their optimality properties.

Definition 4.2 (Efficiency of an Estimator)**.** *The efficiency of an unbiased estimator $\hat{\theta}$ is:*

$$eff(\hat{\theta}) = \frac{1/[nI(\theta)]}{Var(\hat{\theta})}$$

An estimator is called efficient if its efficiency equals 1, meaning it achieves the Cramér-Rao lower bound.

Efficiency provides a standardized measure of estimator quality that allows comparison across different estimation problems. An efficiency of 0.8, for example, indicates that the estimator achieves 80

4.2 Asymptotic Efficiency and Maximum Likelihood Estimation

While few estimators achieve the Cramér-Rao bound exactly in finite samples, many important estimators achieve this bound asymptotically, providing optimal large-sample performance.

Definition 4.3 (Asymptotic Efficiency)**.** *An estimator sequence $\{\hat{\theta}_n\}$ is asymptotically efficient if:*

$$\lim_{n\to\infty} n \cdot Var(\hat{\theta}_n) = \frac{1}{I(\theta)}$$

Theorem 4.2 (Asymptotic Properties of Maximum Likelihood Estimators)**.** *Under regularity conditions, the maximum likelihood estimator $\hat{\theta}_{MLE}$ satisfies:*

1. *Consistency:* $\hat{\theta}_{MLE} \xrightarrow{P} \theta$
2. *Asymptotic normality:* $\sqrt{n}(\hat{\theta}_{MLE} - \theta) \xrightarrow{D} N(0, I^{-1}(\theta))$
3. *Asymptotic efficiency:* $\lim_{n\to\infty} n \cdot Var(\hat{\theta}_{MLE}) = I^{-1}(\theta)$

These properties establish maximum likelihood estimation as a fundamental approach to parameter estimation, providing estimators that are optimal in large samples under mild regularity conditions. The asymptotic normality result also enables the construction of confidence intervals and hypothesis tests based on the normal approximation.

5 Robustness and Breakdown Points

5.1 Robustness Against Model Violations

Real-world data often violate the distributional assumptions underlying classical estimators, making robustness a crucial practical consideration. Robust estimators maintain good performance even when these assumptions are violated, providing reliability in the presence of outliers, heavy tails, or other departures from idealized models.

Definition 5.1 (Robustness)**.** *An estimator is robust if small deviations from the assumed model result in only small changes in the estimator's performance characteristics.*

This informal definition can be made precise through various mathematical frameworks, including influence functions, breakdown points, and minimax approaches. The choice of robustness criterion depends on the specific types of model violations that are of concern in the application.

Definition 5.2 (Breakdown Point)**.** *The breakdown point of an estimator is the smallest fraction of contaminated observations that can make the estimator arbitrarily bad.*

The breakdown point provides a simple but important measure of robustness, quantifying the proportion of outliers that an estimator can tolerate before its performance degrades catastrophically.

Example 5.1 (Breakdown Points of Location Estimators)**.** *Consider estimators of the center of a distribution:*

1. *Sample mean: breakdown point* $= 1/n$ *(a single extreme outlier can make the estimate arbitrarily large)*
2. *Sample median: breakdown point* $= 1/2$ *(up to 50% of observations can be outliers)*
3. *Trimmed mean: breakdown point* $= \alpha$ *(where* α *is the trimming proportion)*

The sample median achieves the maximum possible breakdown point of 50

5.2 M-Estimators and Robust Estimation

M-estimators provide a general framework for robust estimation by replacing the squared loss function used in least squares with more robust loss functions that are less sensitive to outliers.

Definition 5.3 (M-Estimator)**.** *An M-estimator $\hat{\theta}$ minimizes:*

$$\sum_{i=1}^{n} \rho(X_i - \theta)$$

where ρ is a loss function that determines the robustness properties of the estimator.

Different choices of ρ lead to different estimators with varying robustness and efficiency properties:

1. Squared loss: $\rho(x) = x^2$ (sample mean, not robust)
2. Absolute loss: $\rho(x) = |x|$ (sample median, highly robust)
3. Huber loss: $\rho(x) = \begin{cases} x^2/2 & |x| \leq k \\ k|x| - k^2/2 & |x| > k \end{cases}$ (compromise between efficiency and robustness)

The Huber loss function provides a compromise between the efficiency of squared loss for normal data and the robustness of absolute loss against outliers, making it particularly useful in practice.

6 Sufficiency and Optimal Estimation

6.1 Sufficient Statistics and Data Reduction

The concept of sufficiency addresses the question of how much the data can be compressed without losing information about the parameter of interest. Sufficient statistics provide a principled approach to data reduction that preserves all relevant information for parameter estimation.

Definition 6.1 (Sufficient Statistic)**.** *A statistic $T(X_1, \ldots, X_n)$ is sufficient for parameter θ if the conditional distribution of the sample given T does not depend on θ.*

Sufficiency captures the intuitive notion that the statistic T contains all the information about θ that is available in the sample. Once T is known, the specific values of the individual observations provide no additional information about θ.

Theorem 6.1 (Factorization Theorem)**.** *A statistic T is sufficient for θ if and only if the likelihood function can be factored as:*

$$L(\theta; x_1, \ldots, x_n) = g(T(x_1, \ldots, x_n), \theta) \cdot h(x_1, \ldots, x_n)$$

where g depends on the data only through T and h does not depend on θ.

The factorization theorem provides a practical method for identifying sufficient statistics by examining the form of the likelihood function. The factorization reveals which aspects of the data are relevant for inference about θ and which are irrelevant.

6.2 Completeness and Minimum Variance Unbiased Estimation

The combination of sufficiency and completeness leads to optimal unbiased estimators through the Lehmann-Scheffé theorem, which characterizes minimum variance unbiased estimators (MVUEs).

Definition 6.2 (Complete Statistic)**.** *A statistic T is complete for parameter θ if for any measurable function g:*

$$E_\theta[g(T)] = 0 \text{ for all } \theta \Rightarrow P_\theta(g(T) = 0) = 1 \text{ for all } \theta$$

Completeness is a technical condition that ensures the uniqueness of unbiased estimators based on the sufficient statistic. It prevents the existence of non-trivial unbiased estimators of zero, which would complicate the identification of optimal estimators.

Theorem 6.2 (Rao-Blackwell Theorem)**.** *Let $\hat{\theta}$ be an unbiased estimator of θ and let T be a sufficient statistic for θ. Define:*

$$\hat{\theta}^* = E[\hat{\theta}|T]$$

Then $\hat{\theta}^*$ is also unbiased for θ and $Var(\hat{\theta}^*) \leq Var(\hat{\theta})$.

The Rao-Blackwell theorem shows that any unbiased estimator can be improved by conditioning on a sufficient statistic, providing a systematic method for constructing better estimators.

Theorem 6.3 (Lehmann-Scheffé Theorem)**.** *If T is a complete and sufficient statistic for θ, and $\hat{\theta}$ is an unbiased estimator of θ that depends only on T, then $\hat{\theta}$ is the unique minimum variance unbiased estimator (MVUE) of θ.*

This theorem provides the theoretical foundation for identifying optimal unbiased estimators when they exist. The MVUE achieves the smallest possible variance among all unbiased estimators, making it the best choice when unbiasedness is required.

7 Applications in Modern Data Science

7.1 Bias-Variance Tradeoff in Machine Learning

The bias-variance decomposition provides the theoretical foundation for understanding many fundamental concepts in machine learning, including overfitting, underfitting, and the effectiveness of regularization techniques.

In supervised learning, the prediction error on new data can be decomposed into bias, variance, and irreducible noise components. Models with high capacity (many parameters) tend to have low bias but high variance, while models with low capacity tend to have high bias but low variance. The optimal model complexity minimizes the sum of these components.

Regularization techniques such as ridge regression and LASSO explicitly introduce bias to reduce variance, often leading to improved prediction performance. The regularization parameter controls the bias-variance tradeoff, with larger values producing more bias but less variance.

7.2 Robust Methods in Data Science

Modern datasets often contain outliers, measurement errors, or other departures from idealized distributional assumptions, making robustness a crucial consideration in practical data analysis. Robust estimators provide reliable performance across a range of data conditions.

In regression analysis, robust methods such as M-estimators and least absolute deviations provide alternatives to ordinary least squares that are less sensitive to outliers. These methods are particularly valuable in applications where data quality cannot be guaranteed or where outliers may represent important but rare phenomena.

In machine learning, ensemble methods such as random forests and boosting can be viewed as robust approaches that reduce sensitivity to particular features or observations by averaging across multiple models or subsamples.

7.3 High-Dimensional Estimation

Modern data science applications often involve high-dimensional settings where the number of parameters is comparable to or larger than the sample size. In these settings, traditional unbiased estimators may perform poorly due to high variance, making biased but lower-variance alternatives attractive.

Shrinkage estimators, which pull parameter estimates toward a central value, can substantially improve performance in high-dimensional settings. The James-Stein estimator demonstrates that the sample mean can be improved upon even for estimating normal means, challenging the optimality of unbiased estimation in multivariate settings.

Sparse estimation methods such as LASSO implicitly assume that many parameters are zero, introducing bias for non-zero parameters but potentially achieving much lower variance by setting many estimates exactly to zero.

8 Summary

The properties of estimators provide the mathematical foundation for evaluating and comparing different approaches to parameter estimation in data science applications. Understanding bias, variance, consistency, efficiency, and robustness is essential for making informed decisions about estimation methods and interpreting their results correctly.

The bias-variance decomposition reveals the fundamental tradeoff underlying many machine learning concepts and provides theoretical justification for regularization and model selection techniques. Consistency ensures that estimators improve with larger sample sizes, while efficiency measures how well estimators perform relative to theoretical limits.

Robustness considerations are crucial for real-world applications where distributional assumptions may be violated. The theory of sufficient statistics and completeness provides a framework for identifying optimal estimators when they exist, while M-estimators and other robust methods provide practical alternatives when classical assumptions fail.

These theoretical concepts directly inform the design and evaluation of modern data science methods, from machine learning algorithms to statistical models. Understanding these foundations enables practitioners to make principled choices about estimation procedures and to understand their limitations and assumptions.

9 Exercises

1. **Bias-Variance Decomposition:** For the ridge regression estimator $\hat{\boldsymbol{\beta}}_{\text{ridge}} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$, derive expressions for the bias and variance. Show how the bias-variance tradeoff depends on the regularization parameter λ.

2. **Fisher Information Calculation:** Calculate the Fisher information for the exponential distribution with parameter λ: $f(x;\lambda) = \lambda e^{-\lambda x}$ for $x \geq 0$. Verify that the maximum likelihood estimator achieves the Cramér-Rao lower bound.

3. **Consistency Proof:** Prove that the method of moments estimator for the parameter p of a Bernoulli distribution is consistent. Use both the MSE approach and direct application of the weak law of large numbers.

4. **Robustness Analysis:** Compare the breakdown points and asymptotic variances of the sample mean, sample median, and 10

5. **Sufficient Statistics:** For the normal distribution $N(\mu, \sigma^2)$ with both parameters unknown, find a sufficient statistic using the factorization theorem. Show that this statistic is complete.

6. **Rao-Blackwell Improvement:** Starting with the estimator $\hat{\theta} = X_1$ for the mean μ of a normal distribution, use the Rao-Blackwell theorem to construct an improved estimator. Calculate the variance reduction achieved.

7. **Asymptotic Efficiency:** Show that the sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ is asymptotically efficient for estimating σ^2 in a normal distribution.

8. **M-Estimator Properties:** For the Huber M-estimator with loss function $\rho_k(x) = \begin{cases} x^2/2 & |x| \leq k \\ k|x| - k^2/2 & |x| > k \end{cases}$, derive the estimating equation and discuss how the choice of k affects the bias-variance-robustness tradeoff.